

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

EE/EX-4005-CBGS

B.E. IV Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Power System - I

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Sketch neat diagram.

1. a) Give the structure and components of power network. 7
b) What are the various factors involved in power plant economics such as load factor, demand factor, diversity factor, utilization factor? 7
2. a) What are the various constants used in overhead transmission line? 7
b) Write about Ferranti effect with the help of phasor diagram. 7
3. a) Write about Corona formation and explain its advantages and disadvantages. 7
b) Explain about 'skin effect' and proximity effect in reference to overhead lines. 7
4. a) An overhead line has the following data:
Span length $\rightarrow$ 160m, Conductor diameter = 0.95cm,
Weight per unit length of the conductor = 0.65 kg/m,
Ultimate stress $\rightarrow$ 4250 kg/cm²,
Wind pressure $\rightarrow$ 40 kg/km² of the projected area, Factor of safety $\rightarrow$ 5. Calculate the sag. 7

EE/EX-4005-CBGS

PTO

[2]

- b) What are the types of insulator for overhead transmission line failure of insulator? 7
- 5. a) Discuss the effect of increase in voltage on the transmission line efficiency. 7
- b) Explain why the voltage drop is of primary concern in the design of the distributors. 7
- 6. a) Give comparison between isolated and interconnected power system. 7
- b) Prove that transposition of conductors of a three phase transmission line yields equal inductance per phase. 7
- 7. a) A three phase equilaterally spaced transmission line has a total corona loss of 53 kW at 106kV and a loss of 98kW at 110.9kV. What is the disruptive critical voltage between lines? What is corona loss at 113kV? 7
- b) Draw the phasor diagram of a long transmission line and derive an expression for voltage regulation. 7
- 8. Write short notes on the following. 14
 - a) Nominal T and Nominal π representation of transmission line
 - b) String chart and sag template
 - c) Concept of GMR and GMD
